

Temperature-dependent energy gain of bifacial PV farms: A global perspective

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HIGHLIGHTS

- Temperature-dependent energy-yield (YY) and LCOE favor bifacial over monofacial PV.
- An irradiance criterion defines the necessary condition for positive bifacial gain.
- T-dependent vs. STC energy-yield change: (−7, 12)% for Al-BSF and (−5, 5)% for SHJ.
- Global bifacial energy gain for SHJ (bi) vs. Al-BSF (mono) ranges from 12 to 45%.
- YY saturates at elevation ~2 m with a global gain of 0–20% over unelevated farms.

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ABSTRACT

Bifacial solar panels are perceived to be the technology of choice for next-generation solar farms for their increased energy yield at a marginally increased cost. As the bifacial farms proliferate around the world, it is important to investigate the role of temperature-dependent energy-yield and the levelized cost of energy (LCOE) of bifacial solar farms relative to monofacial farms, stand-alone bifacial modules, and various competing bifacial technologies. In this work, we integrate existing irradiance and light collection models with experimentally validated physics-based temperature-dependent efficiency models to compare the energy yield and LCOE of various bifacial technologies across the world. We find that temperature-dependent efficiency changes the energy yield and LCOE by approximately −10 to 15%. Indeed, the results differ significantly depending on the location of the farm (defines the illumination and ambient temperature), the elevation of the module (increases light collection), as well as the temperature-coefficients of various bifacial technologies. The analysis presented in this paper will allow us to realistically assess location-specific relative advantage and economic viability of the next generation bifacial solar farms.

1. Introduction

The PV industry is actively developing bifacial module technology to reduce the levelized cost of energy (LCOE) of the utility-scale solar PV farms [1–12]. Bifacial modules collect light from both faces and the corresponding higher energy, within the same footprint as monofacial modules, makes it a compelling technology for the solar PV industry. A recent literature review by Guerrero-Lemus et al. [13] explains how bifacial technology has developed from its infancy in the 1960s to practical applications and scalability. Despite the possibility of higher output from the same footprint, bifacial PV technology had to wait until

recent innovations for accelerated deployment. The International Technology Roadmap for Photovoltaic (ITRPV) predicts that the worldwide market share for bifacial technology will increase from 15% in 2020 to 40% by 2028 [14]. Since bifacial solar PV farm deployments are now increasing, it is important to understand how module-level bifacial gain translates to location and technology-specific energy yield and LCOE of bifacial farms. Specifically, a realistic assessment of the relative merits of emerging bifacial PV technologies must account for efficiency losses from the thermal effects of the bifacial solar panels installed in utility-scale solar farms. Other spectral and angle-of-incidence effects also affect module efficiency during the early morning

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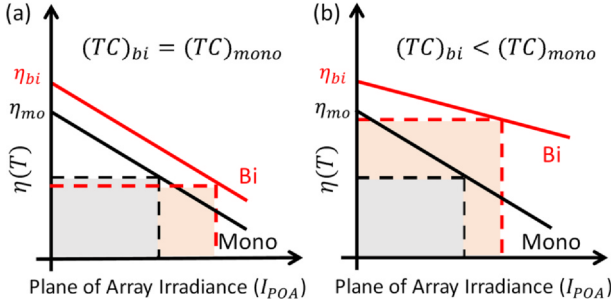


Fig. 1. Effect of temperature coefficient (TC) and irradiance on efficiency of monofacial (black) and bifacial (red) solar panels. Note that energy yield is a product of Plane of Array irradiance (P_{POA}) and efficiency. Hence, higher energy yield requires higher efficiency and higher P_{POA} . (a) Same PV technology; (b) different PV technologies.

and late evening hours of the day. However, these effects are diminished during the operating hours of the PV farms which contribute to energy production.

Fig. 1 shows a linear relation between efficiency and irradiance (for typical operating conditions) and summarizes the importance of temperature-aware performance modeling of solar cells. The output power density (P_{out}) is determined by the temperature-dependent efficiency ($\eta(T)$) and the input irradiance (I_{POA}), i.e., $P_{out} = \eta(T)I_{POA}$. Therefore, the temperature-dependent performance of various technologies can be compared simply by comparing the rectangular boxes in Fig. 1. Since the cell temperature is a function of the ambient temperature T_{amb} , input power I_{POA} , environmental conditions (such as wind speed, relative humidity, and sky temperature), mounting configuration and module construction, and technology-specific thermal coefficient of maximum power, TC , i.e., $T_{cell} = f(T_{amb}, I_{POA}, TC, \dots)$, the conclusions reached regarding bifacial gain based on temperature-independent efficiency, η_{STC} , [15] may not translate into practice. Indeed, the bifacial gain may be negative or positive depending on the geographical location that determines T_{amb} and I_{POA} as well as PV technology which affects the TC and the efficiency gain ($\eta_g = (\eta_{bi}/\eta_{mo})_{STC}$), where η_{mo} is the efficiency of the monofacial cells while η_{bi} is the normalized output of bifacial cells [16,17]. If the bifacial and monofacial panels have the same temperature coefficients (i.e. $(TC)_{bi} = (TC)_{mono}$, as in Fig. 1(a)), higher irradiance would lead to additional heating of bifacial panels, leading to higher degradation of efficiency. Thus, for the limiting (and hypothetical) extreme case of albedo, $R_A = 1$, bifacial panels would outperform monofacial panels only when $I_{POA} < (2\eta_g - 1)/(k(TC)(4\eta_g - 1))$ where k is a location-specific constant and η_g is the bifacial-to-monofacial

efficiency gain. The general condition for arbitrary R_A is derived in Eq. (S5) (see Appendix S1). On the other hand, when bifacial panels have a lower TC as compared to the monofacial panels (Fig. 1(b)), even higher irradiance would not degrade the efficiency below that of monofacial panels. Thus, bifacial PV panels have an optical advantage of greater light collection over monofacial panels but the thermal loss at higher temperatures of operation may potentially negate some of the optical gains. In short, temperature-dependence fundamentally alters the conclusions regarding location-specific bifacial energy gain across the world.

Recent studies have therefore focused on the temperature-dependent performance of stand-alone monofacial and bifacial PV modules [18–23]. For example, Refs. [18–20] contain systematic and detailed analyses of temperature coefficients for different PV materials and technologies. The open rear surfaces of a bifacial cell increase power-input due to albedo and power-loss due to IR radiation. In principle, the bifacial cells can operate cooler than a monofacial cell [22]. Several theoretical studies have explored the temperature-dependence of several performance parameters of solar cells as well as solar PV systems [21,23]. Lopez-Garcia et al. have experimentally (indoor and outdoor) analyzed the temperature coefficients (TC) of bifacial c-Si PV modules. They concluded that bifacial TC is generally significantly lower than monofacial TC [24]. Recent studies [25–28] based on a coupled optical-electrical-thermal model of the bifacial photovoltaic module showed a 22% bifacial gain in yearly energy yield and suggested tilt and elevation optimizations for stand-alone bifacial modules.

Compared to the extensive literature for stand-alone modules, there are only a few reports of temperature-dependent loss of energy yields of various monofacial and bifacial technologies at the farm level [15,29,30]. The standalone module and a farm differ significantly in terms of albedo collection related to the row-to-row mutual shading, leading to substantial changes in self-heating. In comparison to Ref. [15], this paper includes two important generalizations: (1) Increase of the energy-yield with elevation and (2) the decrease in energy yield with temperature. Compared to the previous analysis in Ref. [15], our key conclusion regarding the general location-dependent advantage of bifacial solar modules over monofacial modules remain the same. However, our new work shows that optimum elevation and lower temperature coefficient make bifacial cells attractive for places that we previously thought would only offer marginal improvement.

In this paper, we model a generalized bifacial solar PV farm that includes the effects of temperature-dependent efficiency, as portrayed in Fig. 2. The integrated model will combine optical, thermal, and electrical models with economic analysis to estimate the energy yield and LCOE of a solar PV farm. The effects of temperature on a module

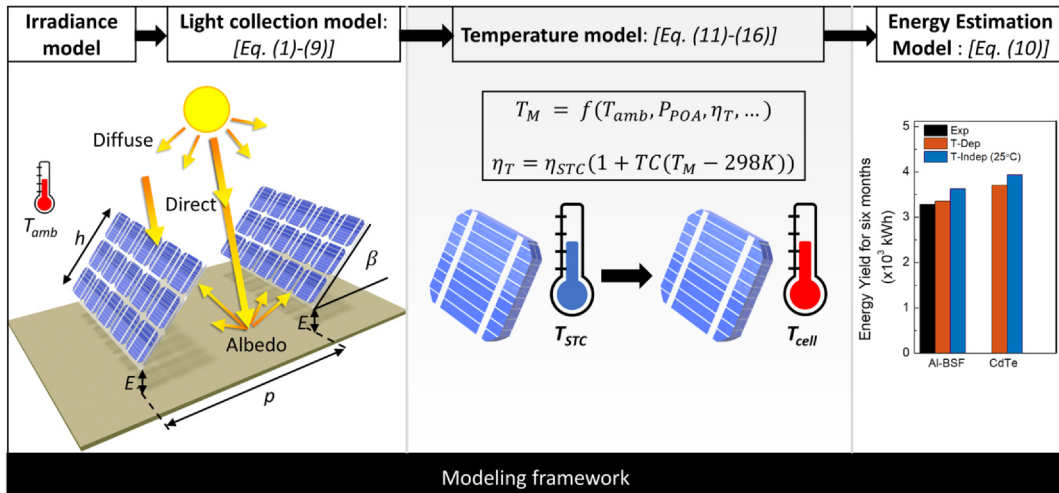


Fig. 2. Flow diagram of the modeling framework with physical parameters for temperature-dependent energy yield estimation.

Table 1

Equations associated with Optical Model: irradiance modeling and light collection. Note that Eqs. (1)–(9) are time-dependent while Eq. (10) is an integral over time (a year).

$$\begin{aligned}
 I_{GHI} &= I_{DNI} \times \cos(\theta_z) + I_{DHI} \quad (1) \\
 I_{PV:DNI}^{F, Farm} &= I_{DNI} \cos \theta_F (1 - R(\theta_F)) \eta_{dir} \quad (2) \\
 I_{PV:DHI}^{F, Farm} &= I_{DHI} \eta_{diff} / h \int_0^h \frac{1}{2} (1 + \cos(\psi(E, l) + \beta)) dl \quad (3) \\
 I_{PV:Alb}^{F, Panel} (l) &= I_{Gnd:DNI} \times R_A \times F_{dl-gnd}(E, l) \times \eta_{diff} \quad (4) \\
 I_{PV:Alb, dir}^{F, Farm} (l) &= 1/h \int_0^h I_{PV:Alb, dir}^{F, Panel}(E, l) dl \quad (5) \\
 I_{PV:Alb, diff}^{F, Panel} (l) &= I_{Gnd:DHI} \times R_A \times F_{dl-gnd}(E, l) \times \eta_{diff} \quad (6) \\
 I_{PV:Alb, diff}^{F, Farm} &= 1/h \int_0^h I_{PV:Alb, diff}^{F, Panel}(E, l) dl \quad (7) \\
 I_{PV:Alb}^{F, Farm} &= I_{PV:Alb, dir}^{F, Farm} + I_{PV:Alb, diff}^{F, Farm} \quad (8) \\
 I_{PV(T)}^{F, Farm} &= I_{PV:DNI}^{F, Farm} + I_{PV:DHI}^{F, Farm} + I_{PV:Alb}^{F, Farm} \quad (9) \\
 YY_T(p, \beta, h, E, \gamma_A, R_A) &= \int_0^1 I_{PV(T)}^{F, Farm}(p, \beta, h, E, \gamma_A, R_A) dY \quad (10)
 \end{aligned}$$

and farm at different locations over the globe are compared and it is found that the *stand-alone* modules are affected by temperature more than the *farms*. Moreover, the roles of temperature coefficients of different materials (e.g., Aluminum Back Surface Field (Al-BSF) and Silicon Heterojunction (SHJ)) are explored in terms of efficiency degradation and bifacial gain (bifacial vs. monofacial). Lower temperature coefficient ($0.26\%/^{\circ}\text{C}$) for SHJ (bifacial) shows reduced efficiency degradation and yields higher bifacial gain (**20–40% globally**) compared to Al-BSF (monofacial) which has a higher temperature coefficient ($0.41\%/^{\circ}\text{C}$). Next, combined dependencies of temperature and elevation on the energy yield are considered and we find that the energy yield saturates after a threshold value of elevation (typically, $E_0 = 2$ m). About **1–15%** increase in energy yield between elevated ($E = 2$ m) and unelevated ($E = 0$ m) bifacial farms is observed around the world.

In Section 2 we describe the optical, thermal, and electrical models used in our study, and the associated equations are organized in Tables 1 and 2. In Section 3 we show the results and discuss important trends and features. We vary several design parameters of a solar PV farm e.g., elevation, PV technology, bifacial/monofacial, and TC , to quantify the effects of temperature-dependent efficiency on energy yield, LCOE, and optimum design (tilt angle) of the farm. Finally, in Section 4, we summarize the study and conclude the paper. Note that the symbols used in this paper are defined in Table 3.

2. Modeling and validation

An LCOE-aware modeling of bifacial solar farm involves the calculation of irradiance, collection of sunlight, and an electrical output coupled to economic analysis. The optimized design (panel orientation, tilt, elevation, and spacing) of the farm then can be evaluated in terms of energy or LCOE.

Consider a solar farm oriented at an azimuth angle γ_A from the North, with panels of height h , tilted at an angle β , elevated from the ground to E , and with pitch p over a ground of albedo R_A . In the following discussions, we will focus on three aspects of the modeling framework for the specified solar farm: (A) Re-parameterizing LCOE, (B) Irradiance modeling, and (C) Collection of light. These topics are discussed below.

2.1. Farm model with LCOE*

LCOE is defined as the ratio of the total cost of a PV system to the total energy yield of the system over its lifetime [31], i.e.,

$$\begin{aligned}
 LCOE &= \frac{\text{Total Cost (\$)}}{\text{Total Energy Yield (kWh)}} \\
 &= \frac{C_{sys}(Y=0) + (\sum_{k=1}^Y C_{om}(k)) - C_{rv}(Y)}{E(Y)} \quad (17)
 \end{aligned}$$

where $C_{sys}(Y=0)$ is the initial fixed installation cost of the system (i.e., at $Y=0$). $C_{sys}(Y=0)$ includes the cost of modules ($c_{m,0}$), the cost of land ($c_{l,0}$), and the balance of system cost ($c_{bos,0}$) such as labor, permit, racks, inverters, etc. The recurring operations and maintenance cost (C_{om}) scales with the cost of maintaining individual modules ($c_{om,m}$) and the cost of maintaining the land ($c_{om,l}$). Finally, C_{rv} is the residual value of the modules ($c_{rv,m}$), the land ($c_{rv,l}$), and the equipment to be regained when the farm is decommissioned. C_{om} and C_{rv} are a function of the lifetime (number of years, Y) for which the solar farm is operated.

Since the costs vary with the number and size of PV modules and the solar-farm land, the LCOE expression can be described using the dimensions of a solar farm and the modules installed, as shown in Eq. (18) below.

$$LCOE = \frac{M(r) \cdot h \cdot M \cdot Z + L(r) \cdot p \cdot M \cdot Z + C_{bos,f}}{YY_T(p, h, E, \beta, \gamma_A, R_A) \cdot M \cdot Z \cdot h \cdot \chi(d, r)} \quad (18)$$

Here M is the cost per unit meter of the module (height), L is the cost per unit meter of land (pitch), M is the number of rows/arrays of modules, and Z is the number of modules in an array (in the z -direction, into the page). $YY_T(=E_0)$ is the yearly energy yield per meter of a pristine module for one period/pitch (p) such that the yearly energy of the farm is $E(Y) = YY_T \cdot M \cdot Z \cdot h \cdot \chi(d, r)$, where $\chi = \sum_{k=1}^Y (1-d)^k (1+r)^{-k}$. YY_T is a function of the physical design parameters ($p, h, E, \beta, \gamma_A, R_A, T$). The lifetime (Y , typically 25 years) of a solar farm is defined as the time duration before the performance (efficiency) of a solar farm degrades by 20%. Thus, the degradation rate (d , typically $0.7\%/year$) defines the lifetime of a solar farm. The discount rate (r) accounts for the devaluation of predicted future earnings.

Further, the cost associated with the balance of system ($c_{bos,f}$) is typically negligible as compared to the essential costs, C_M and C_L [4,32], and does not affect the design optimization of the farm. With these considerations, we arrive at the ‘essential leveled cost of energy’ ($LCOE^*$), as follows:

$$LCOE^* = \frac{LCOE \cdot \chi}{L} = \frac{M/L + p/h}{YY_T} = \frac{p/h + M_L}{YY_T} \quad (19)$$

Here, $M_L(= M/L)$ is the ratio of the cost of module per unit length (height) to the cost of land per unit length (pitch). M_L essentially captures the costs of a solar farm whereas p/h and $YY_T(p, h, E, \beta, \gamma, R_A)$ contains information about the physical parameters of the farm. M_L varies with the technology and location in the world. In this paper, we assume a typical value of $M_L = 15$ [15]. Given

Table 2

Equations associated with Thermal Model.

$$\begin{aligned}
 \eta(T_{cell}) &= \eta_{STC} (1 - TC(T_{cell} - T_{STC})) \quad (11) \\
 T_{cell} &= T_{amb} + I_{POA} \times e^{a+(b \times WS)} + \Delta T \times \frac{I_{POA}}{1000} \quad (12) \\
 T_{cell} &= T_{amb} + c_T \frac{\tau_a}{u_L} I_{POA} \left(1 - \frac{\eta(T_{cell})}{\tau_a} \right) \quad (13) \\
 T_{cell} &= T_{amb} + (I_{POA} - \eta I_{POA} - \gamma I_{POA}(subband)) \times \frac{F}{1000} \quad (14) \\
 T_{cell} &= T_{amb} + (I_{POA,top} - \eta_{top} \times I_{POA,top} + I_{POA,bot} - \eta_{bot} \times I_{POA,bot} - I_{POA}(subband)) \times \frac{F}{1000} \quad (15) \\
 I_{POA} &= I_{PV:DNI}^{F, Farm} / \eta_{dir} + (I_{PV:DHI}^{F, Farm} + I_{PV:Alb}^{F, Farm}) / \eta_{diff} \quad (16)
 \end{aligned}$$

Table 3
Glossary of the symbols used in this paper.

Parameters	Definition
TC	Temperature Coefficient (%/°C)
T_a	Ambient Temperature (°C)
T_M	Module Temperature (°C)
P_{in}/P_{out}	Input Optical Power Collected by the Panel / Output Electrical Power (W/m ²)
η	Power conversion efficiency (%)
I /Irradiance	Input optical power over the plane of array (W/m ²)
a,b	Fitting parameters in the King's model (no unit, s/m)
WS	Wind Speed (m/s)
c_T	Correction term used for the daily average temperature data (1, in this paper)
τ	Transmittance of glazing (0.95, in this paper)
α	Absorbed fraction (0.8, in this paper)
u_L	Heat loss coefficient (21.5 W/m ² K, in this paper)
γ	Coefficient of sub-band power contribution to heating (No unit)
θ_Z	Zenith Angle (degrees)
θ_F	Angle of incidence at the front face of the panel
Pitch (p)	Row-to-row distance between the bottom edges of consecutive arrays (m)
Height (h)	Height of the panel (1 m, in this paper)
E	Elevation
β	Tilt angle
γ_A	Azimuth angle from measure from the North
R_A	Albedo (0.5, in this paper)
C/	Cost / Cost per unit meter
M_L	Module to land cost ratio (15, in this paper)
M	Number of rows/arrays of modules
Z	Number of modules in an array
d	Yearly degradation rate in energy conversion (0.7%, in this paper)
Y	Lifetime of a farm (25 years, in this paper)
YY	Yearly Yield
r	Discount rate
F	Fitting parameter in Alam-Sun model which accounts for module assembly and its related effects (like transmittance, glazing, heat loss coefficient)
Sub/Super-scripts	Definition
a	Ambient
Alb	Albedo
bos	Balance of system
bot	Back side
DHI	Diffuse Horizontal Irradiance
diff	Diffuse
dir	Direct
dl	Differential element along the height of a panel
DNI	Direct Normal Irradiance
F	Front
f	Fixed cost
Farm	PV Farm
GHI	Global Horizontal Irradiance
Gnd	Ground
in	Input
L/L	Land
M, cell	cell/module
om	Operation and maintenance
Panel	PV panel
PV	Photovoltaic
rv	Residual value
STC	Standard testing condition
sys	System
top	top side
Total	Total
Z	Zenith

M_L , we can perform the physical design optimization of a solar farm to find a minimum $LCOE^*$. Eq. (19) shows that $LCOE$ is proportional to $LCOE^*$, since L and χ are location-specific constants. Therefore, minimizing $LCOE$ is equivalent to minimizing $LCOE^*$ for a given location. Ref. [15] provides a more detailed derivation and discussion on $LCOE$ and $LCOE^*$.

Next, we will estimate the amount of sunlight falling on a farm at

any location in the world.

2.2. Irradiance modeling and light collection

The energy yield of a bifacial solar farm for a given albedo is numerically modeled in three steps.

- (1) **Irradiance model.** We calculate the of sunlight incident at a location defined by its latitude and longitude. This requires the Sun's trajectory (zenith (θ_Z) and azimuth angle (γ_A)) and the irradiance [29]. The Global Horizontal Irradiance (GHI or I_{GHI}) is ideally given by Haurwitz clear sky model [33,34]. We renormalize this irradiance based on the NASA Surface meteorology and Solar Energy database [35] to find the local variation in GHI. The GHI is split into direct light (DNI or I_{DNI}) and diffuse light (DHI or I_{DHI}) using Orgill and Hollands model [36].
- (2) **Light collection on panels.** We quantify the amount of light collected by the solar panels installed at that location. The panels have height h , tilted at an angle β , separated by pitch (or period) p , and are oriented at an array azimuth angle $\gamma_A = 180^\circ$ (i.e., south-facing panels) for farms in the northern hemisphere and $\gamma_A = 0^\circ$ (i.e., north-facing panels) for farms in the southern hemisphere or $\gamma_A = 90^\circ$ for panels facing E-W direction. The collection of light on panels from the three components of irradiance, i.e., direct, diffuse, and albedo light are formulated separately and analyzed accordingly. Our approach to model the collection of light follows Ref. [15] and the equations are summarized in Table 1. We calculate the angle of incidence (AOI) of light to find the component of direct light (I_{DNI}) falling on the tilted panel's front or the back face (depending on the tilt angle). θ_F is AOI for front face, R is the angle-dependent reflectivity ($R(\theta_F)$) of the panel, $\psi(l)$ is the viewing angle at length l , and F_{dl-gnd} is the view factor from ground to a point on the panel. We extend the ground mounted panel-array model in [15] to include calculations for elevated panels. The pattern of light falling on the ground, estimated by the view factor ($F_{gnd-sky}$), and subsequently the albedo light collected by the panels, estimated by the view factor (F_{dl-gnd}), both vary with the elevation (E) of the panels. A detailed explanation of the albedo light model varying with elevation is presented in the appendix and Refs. [37,38].
- (3) **Energy yield.** We finally find the daily and yearly energy-output of the farm. Using the Eqs. (2), (3), and (8) in Table 1, we arrive at Eq. (9) to find the time-varying spatially distributed light collection on the panels. This information is used in the circuit model to find the equivalent power generation. To estimate energy output, we integrate the power generated over the desired duration of time. We define the energy yield per pitch of a farm over one year as yearly yield (YY) given by Eq. (10). Now, we can vary the albedo to determine the variation in YY.

2.3. Temperature-dependent efficiency models

The temperature-dependent efficiency loss involves a complex interplay of increased absorption due to bandgap reduction vs. increased dark-current and reduced mobility. Within practical illumination intensities, it is well-known that the efficiency $\eta(T)$ scales linearly with the module/cell temperature, see Eq. (11) in Table 2 where the rate/slope of degradation is given by the absolute temperature coefficient (TC) [21,39].

The module/cell temperature further depends on the ambient temperature (T_{amb}), wind speed (W_s), module mounting type, the efficiency of the module, and other factors. In this work, we have considered King (Eq. (12), Table 2), Skoplaki (Eq. (13), Table 2) and Alam-Sun (Eq. (14), Table 2) models to estimate the module/cell temperature based on location-specific the ambient temperature.

Briefly, **King's Model** (Eq. (12)) [40,41] accounts for the wind speed (WS) and cell to module temperature differences (ΔT) where the

parameters 'a' and 'b' are negative and are used as fitting parameters for the module type and deployment. This simple model does not explicitly account for output power (efficiency is absent in the analytical expression) hence, it doesn't need to be solved self-consistently with Eq. (11), Table 2.

Unlike King's model, **Skoplaki's model** (Eq. (13)) [42,43] accounts for the module efficiency. It also includes: (i) experimentally-fitted heat loss coefficient u_L which is found to be 21.5 for our experimental data and (ii) the correction term c_T which accounts for averaging the ambient temperature and irradiance. We found that instantaneous data (every 1-min) does not require any correction term to predict the energy yield. Additionally, c_T increases with the duration of the average, i.e., daily average requires a smaller correction term than the monthly average.

Finally, the **Alam-Sun** (Eq. (14)) model generalizes the Skoplaki's model, which allows us to account for the temperature rise due to the sub-bandgap absorption. The model uses a parameter ($0 < \gamma < 1$) to account for the fraction of sub-bandgap irradiance/power that is not absorbed by the solar module. When $\gamma = 0$, the Alam-Sun model reduces to the Skoplaki model. Sub-bandgap absorption is an important consideration because the bifacial glass-glass modules may have reduced sub-bandgap absorption compared to monofacial glass-backsheet modules. The relatively cool bifacial modules (for the same I_{POA}) would improve the bifacial gain. A slightly generalized version of the Alam-Sun model (Eq. 15) is necessary to account for the asymmetric absorption from the rear side of the module.

These cell temperature estimation models are coupled to the temperature dependence of efficiency and solved self-consistently. Once the temperature-dependent efficiency is obtained, we multiply it with the incident light intensity to deliver the energy output (i.e. the rectangular box in Fig. 1). Fig. 3 demonstrates the validity of the T -dependent simulation model with the experimental data from Sandia National Labs, Albuquerque, NM, USA. We use the experimentally available I_{POA} along with the Skoplaki model to find that the experimentally observed energy yield (Black circles) accurately matches the simulated T -dependent energy yield (Red-solid line). Note that the T -independent efficiency model overestimates the energy yield (Blue-dashed line). The inset in Fig. 3 shows the accuracy of simulated T -dependent energy yield. Further, Fig. 4 summarizes the importance of the sub-bandgap absorption (Alam-Sun model, Eq. (14)) and the effect of $0 < \gamma < 1$ on efficiency. When $\gamma = 0$ (monofacial-like), the sub-bandgap irradiance is absorbed causing the panels to be hotter and efficiency to be lower. On

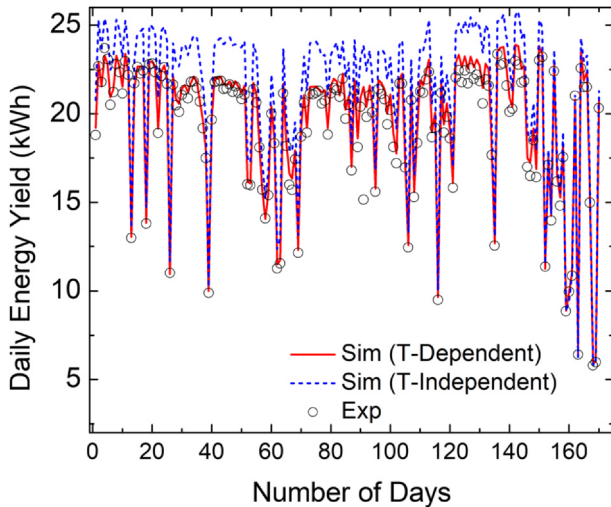


Fig. 3. Measured daily energy yield (open circles) for Albuquerque, NM, USA. Simulation is done based on the nameplate efficiency along with (red solid line) and without (blue dashed line) its temperature dependence. (Inset) A zoomed-in version of Fig. 3 from day 40 to 50.

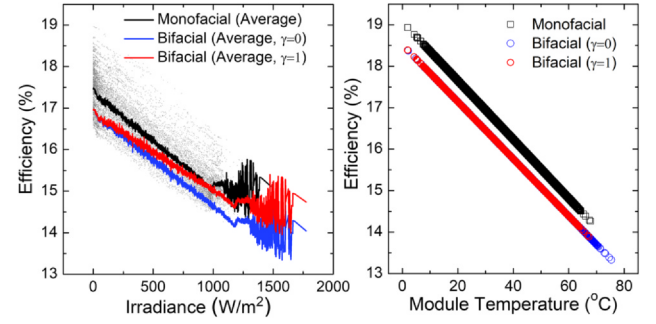


Fig. 4. (a) Average efficiency vs Irradiance for both monofacial and bifacial solar panel where the background symbols (grey small open circles) represent the measured data for monofacial string. (b) Efficiency vs T for monofacial and bifacial solar modules. The TC is same for both the panels which is reflected in the slopes – however, bifacial is cooler (shown as red arrow) and lead to ~ 90 kWh gain when the sub-band spectra is assumed transparent ($\gamma = 1$). The change in the slope indicates the change in the temperature of the panel due to the sub-band transparency.

the other hand, for $\gamma = 1$, the solar panels are completely transparent to sub-bandgap illumination and the panels are cooler with higher efficiency. We observe ~ 90 kWh energy gain for 12 modules for a duration of six months with $\gamma = 1$ as compared to $\gamma = 0$.

2.4. Energy output with temperature dependence

Finally, the power generated by the panels under STC (described in section 2.2) is corrected using the temperature-dependent efficiency model i.e., Eqs. (11), (13), and (16) from Table 2. For energy output, we integrate the power generated over the desired period of time. We define the energy yield per pitch of a farm over one year as yearly yield (YY) as shown below (same as Eq. (10) in Table 1).

$$YY_T(p, \beta, h, E, \gamma, R_A) = \int_0^1 I_{PV(T)}^{Total}(p, \beta, h, E, \gamma_A, R_A) dY \quad (20)$$

Fig. 3 validates our temperature-dependent energy yield prediction with experimental data from Sandia National Laboratory. The deployed panels were manufactured by Canadian Solar (CS6K-275M 275 W mono-Si) with a module efficiency $\eta_{STC} = 16.8\%$, temperature coefficient $TC = 0.41\%/^{\circ}C$, and outer dimensions of 1650×992 mm. The modules are installed S-facing, 35° fixed-tilt at Sandia National Laboratories in Albuquerque, New Mexico USA (35.05° N, 106.54° W) at an elevation of 1663 m above sea level and are mounted in a 2-up landscape fashion, with a row spacing of 4.88 m. They are grid-connected in four strings of 12 modules. Performance data presented in this paper is derived from string-level I-V curves that were measured at 30-minute intervals over a period of ~ 170 days. The T -dependent energy estimation (solid red line) follows the experimental data (black circles) almost exactly while the T -independent energy (dashed blue line) is overestimated.

3. Results and discussion

As the first step of calculating the temperature-dependent energy yield and LCOE, we examine the temperature dependence of monofacial and bifacial module configurations. The module temperature is calculated self-consistently with efficiency and irradiance. Fig. 4 shows the trend in $\eta(T)$ between monofacial and bifacial modules. Although the bifacial modules collect more light (irradiance), the apparent temperature coefficient (slope of the red line in Fig. 4(a)) is relatively lower. This can be explained by the Sun-Alam model, where γ is the fraction of sub-bandgap irradiance/power that is not absorbed by the solar module. Fig. 4(a) shows the effect of spectral irradiance on the modules. Looking at the two extreme cases of $\gamma = 0$ (monofacial with an

opaque back sheet) and $\gamma = 1$ (bifacial with a transparent back sheet), we see that the efficiency vs. irradiance plot has a steeper slope for monofacial case. The rapid decrease in efficiency is a consequence of increased temperature (Fig. 4(b)) due to additional sub-bandgap (or IR) absorption. Since TC is a significant parameter in temperature-dependent efficiency estimation, solar module manufacturers should try to minimize TC through material and process engineering. Note that all the temperature-dependent parameters are denoted by subscript T while temperature-independent parameters are denoted by subscript 0.

3.1. Self-heating affects a stand-alone module more than modules in a farm

A bifacial module in a solar PV farm collects less irradiance (I_{POA}) as compared to a standalone bifacial module due to mutual shading (periodic shading on the ground). Thus, our self-consistent model predicts the temperature of a standalone module to be slightly higher than that of a module in a farm. Note that the temperature difference would change somewhat ($< 3-4^\circ$) once the “heat-island effect” is accounted for [44]. This relatively small correction reflects two counter-balancing effects of ground illumination vs. heat-trapping in a farm vs. in a standalone module. In general, the temperature-dependent efficiency (η_T) and energy yield (YY_T) degradation, with respect to temperature-independent η_{STC} (or η_0) and YY_0 , are more prominent for a standalone module as compared to a farm, see Fig. 5. We observe that the degradation in YY is enhanced for standalone bifacial modules as compared to bifacial farms. In Fig. 5, we consider four specific locations with increasing latitudes, namely, Dubai (24. 5°N, 55. 5°E), Sydney (33. 85°S, 150. 65°E), Washington D.C. (38. 91°N, 77. 04°W), and Stockholm (59. 32°N, 17. 84°E) with decreasing average annual temperatures 34, 22, 20, 11°C (high) and 23, 14, 8, 5°C (low), respectively. For locations with hotter climates the percentage change in YY_T and YY_0 is negative, due to efficiency degradation at higher temperatures. On the other hand, for places with very cold climates, such as Stockholm, the associated percentage change is positive and higher for a farm in comparison to a standalone module. The apparent exception at Stockholm arises due to colder weather i.e., $T_{amb} < T_{STC}$, during most of the year. Negative ambient temperature and low irradiance (I_{POA}) for a farm, leads to lower T_{cell} in a solar farm as compared to T_{cell} in a standalone module, see Eq. (13). Thus, from Eq. (11), $\eta_T(\text{farm}) > \eta_T(\text{module})$ and $\Delta YY = I_{POA} \times \Delta \eta_T$ is higher for a farm as compared to a standalone module.

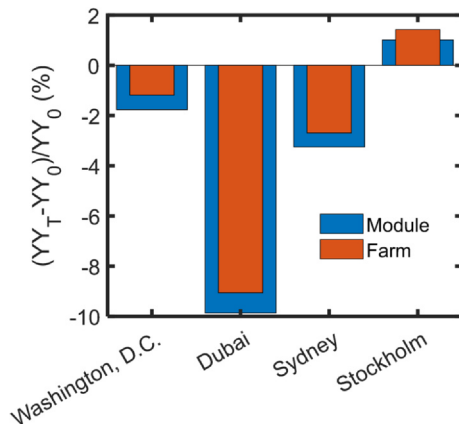


Fig. 5. Comparing module and farm-level η_T vs. η_0 based percentage change in energy yield for different locations around the world. We observe that degradation is higher for an elevated ($E = 2$ m) standalone module as compared to an elevated farm, except for locations at very high latitude e.g., Stockholm.

3.2. Technology and geography dictate the temperature-dependent yield-loss of monofacial vs. bifacial farms

The worldwide maps in Fig. 6 depict the dependence of gain/loss in energy yield between temperature-dependent efficiency (η_T) vs. constant efficiency (η_0 or η_{STC}). Clearly, hotter places like the Sahara Desert, Mexico, and southern India, show a loss in energy yield due to degradation in efficiency with higher temperature as compared to efficiency at STC (η_{STC}). On the other hand, for colder locations like Siberia, Gobi Desert, and northern Canada and Europe, with lower average temperatures, show an improvement in efficiency, with respect to η_{STC} , leading to higher energy yield.

This general trend is quantitatively shown in Fig. 6 (row 1) for monofacial farms: globally, we observe -7% to $+12\%$ of change in temperature-dependent energy yield (YY_T) with respect to energy yield (YY_0) estimated using constant efficiency. Fig. 6 (row 2) shows a lower magnitude of change (-7% to $+10\%$) for bifacial farms. Here we assume the same temperature coefficient ($TC = 0.41\%/^\circ C$) for monofacial and bifacial modules (rows 1 and 2 in Fig. 6) with the same Al-BSF technologies. Since the amount of light collected by the bifacial panels (I_{POA}) is higher than the monofacial panels, the change in cell/module temperature, from Eq. (13) in Table 2, is higher for bifacial modules leading to a higher change in efficiency. However, the normalized output (efficiency) for bifacial modules is higher than the monofacial modules (as shown by the y-intercepts in Fig. 1(a)). Overall, the combination of higher normalized output and higher degradation of efficiency for bifacial modules leads to an approximately similar percentage change in energy yield between temperature-dependent vs. constant efficiencies. Moreover, Fig. 6 (row 3) displays the same quantities for bifacial silicon heterojunction technology which has a lower TC ($= 0.26\%/^\circ C$). Lower TC leads to lesser degradation of efficiency with temperature and eventually the range of change in YY (-5% to $+5\%$) is lower as compared to Al-BSF ($TC = 0.41\%/^\circ C$).

Fig. 6 also displays the trend in relative percentage change between minimum $LCOE^*$ estimated using temperature-dependent efficiency ($LCOE_{min,T}^*$) with respect to $LCOE^*$ calculated using constant efficiency ($LCOE_{min,0}^*$). Since $LCOE^*$ is inversely proportional to the yearly energy yield (YY), hence, the trends are reversed as compared to energy yield. Colder places show a decrease in $LCOE^*$ by 8% , whereas places with hotter climates show a 8% increase for bifacial farms. The change (-8% to $+8\%$) in $LCOE^*$ after temperature correction is significant and cannot be ignored while estimating the levelized cost of energy of a solar PV farm.

The design of solar modules in a farm (represented by the optimum tilt angle and the associated pitch for no mutual shading) does not change significantly. Fig. 6 (maps on the right) shows the difference in the optimum tilt angle when we consider η_T vs. η_0 . This difference varies from -3° to $+3^\circ$ around the globe. Hence, for all practical purposes, the design optimization presented in [15] does not require adjustment even after temperature-dependent efficiencies are included. However, an accurate estimation of YY and $LCOE^*$ is achieved using a more accurate temperature-dependent efficiency model.

3.3. Technology-choice and geographical location determine the Temperature-dependent bifacial gain

Since it has been established that the temperature-dependent model is essential for the correct estimation of YY and $LCOE^*$, we will now explain the bifacial gain. We will compare the Silicon Heterojunction (SHJ) ($TC = 0.26\%/^\circ C$) [45,46] and Aluminum Back Surface Field (Al-BSF) ($TC = 0.41\%/^\circ C$) bifacial panel performance to that of Al-BSF monofacial panels ($TC = 0.41\%/^\circ C$). The global trends in bifacial gains for YY and $LCOE^*$ for temperature-independent efficiency calculations can be found in [15]. Fig. 7 displays the same parameters for temperature-dependent Al-BSF (bifacial) (row 1) and SHJ (bifacial) (row 2) versus Al-BSF monofacial farm calculations. We observe that the

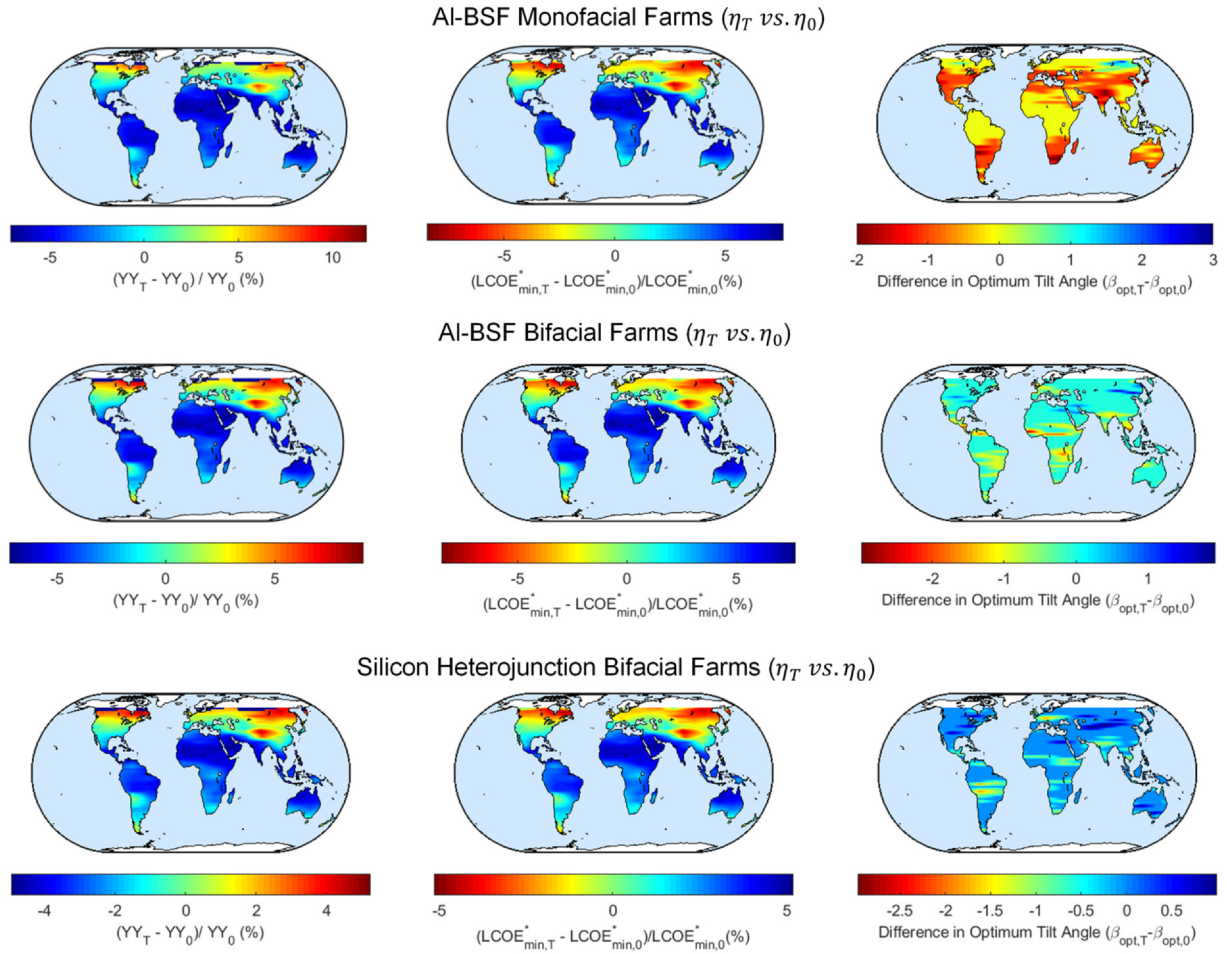


Fig. 6. Global maps comparing yearly energy yield, $LCOE^*$, and difference in optimum array tilt angle for temperature-dependent efficiency (subscript T) and temperature-independent efficiency (subscript 0). Note that $R_A = 0.5$, $E = 0$ m, and $M_L = 15$. (Row 1) Monofacial Farms, e.g., AI-BSF, (Row 2) Bifacial farms with a high temperature coefficient ($TC = 0.41\%/^{\circ}C$), and (Row 3) Bifacial farms with a lower temperature coefficient ($TC = 0.26\%/^{\circ}C$) e.g., Silicon Heterojunction (SHJ) solar cells.

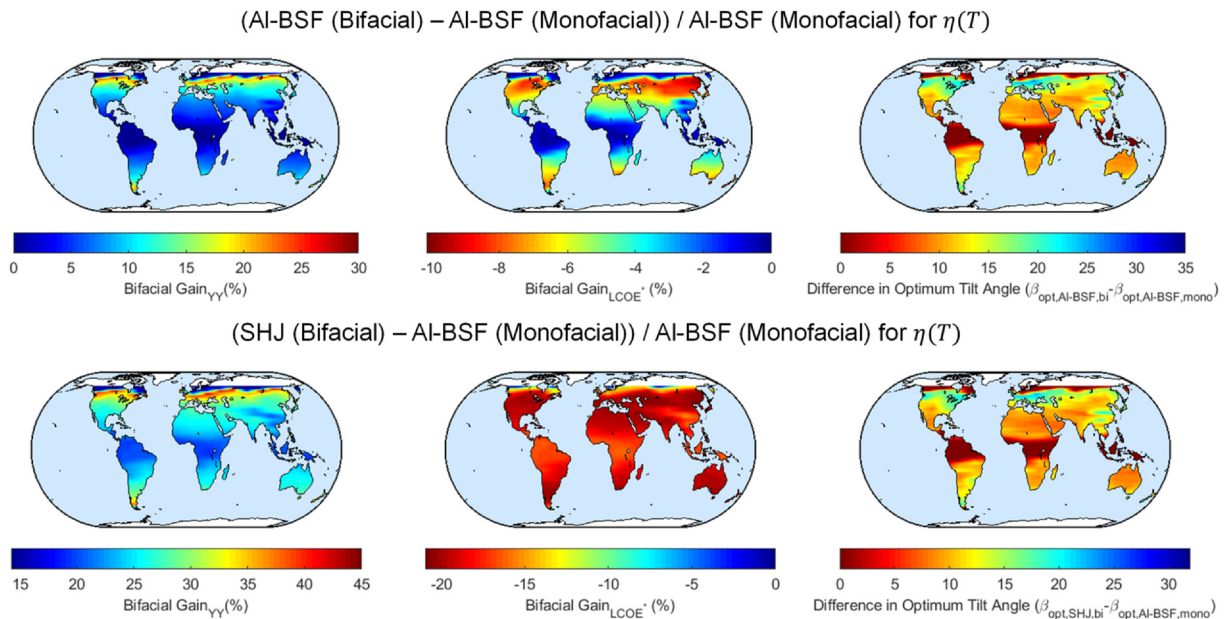


Fig. 7. Global maps comparing bifacial gain in yearly energy yield, $LCOE^*$, and difference in optimum array tilt angle for AI-BSF monofacial farms ($TC = 0.41\%/^{\circ}C$) vs. (Row 1) AI-BSF and (Row 2) Silicon Heterojunction bifacial farms ($TC = 0.26\%/^{\circ}C$). Note that $R_A = 0.5$, $E = 0$ m, and $M_L = 15$.

bifacial gain for the yearly yield of Al-BSF is 0–20% for latitudes $< 50^\circ$ and reaches up to 30% for even higher latitudes. The reduction in $LCOE^*$ for bifacial over monofacial is 0–10% around the world. These percentages are higher than that estimated (2–12%) for temperature-independent (constant) efficiency in Ref. [15]. Calculations comparing SHJ (bifacial) and Al-BSF (monofacial) show a much higher bifacial gain for yearly yield (12–45%) and reduction in $LCOE^*$ for bifacial over monofacial parameters (15–25%) around the world. This is due to the higher temperature coefficient for Al-BSF as compared to SHJ panels leading to larger performance degradation of Al-BSF panels and lower bifacial gain. The relative gain simply reflects the fact that the efficiency of bifacial PV degrades less than monofacial PV at a given ambient temperature.

Recall that in Fig. 1, we conceptually compared monofacial and bifacial farms with different efficiencies and temperature coefficients. The observations shown here clearly align with our hypothesis of enhanced temperature-dependent performance for bifacial panels with higher $\eta_{STC}(SHJ) = 19.7\%$ and lower temperature coefficient ($TC(SHJ) = 0.26\%/^\circ C$) as compared to monofacial panels with lower $\eta_{STC}(Al-BSF) = 16.8\%$ and higher temperature coefficient ($TC(Al-BSF) = 0.41\%/^\circ C$).

Clearly, temperature coefficient (TC) is an important parameter for accurate efficiency and energy estimation and it varies for different PV technologies. Although bifacial modules collect more light, most bifacial technologies have a lower TC than monofacial ones leading to further enhanced bifacial gain. Furthermore, bifacial technologies do not absorb sub-bandgap (IR) radiation leading to lower heating. Fig. 8 portrays this trend for panels deployed at Sandia National Labs, Albuquerque, and validates our hypothesis. Since the temperature coefficient of Al-BSF ($TC = 0.41\%/^\circ C$) is higher than CdTe ($TC = 0.32\%/^\circ C$), the energy yield for the former is lower. Further, for both PERC and SHJ, T-dependent estimation shows higher energy yield for bifacial as compared to monofacial due to lower IR absorption. We saw similar results for the global maps in Fig. 7.

3.4. Temperature-Dependent energy yield of elevated Farms: Self-Heating vs. Improved light collection

The energy yield increases with elevation due to improved collection of albedo light at the back face of the panel. Increased light collection increases the temperature, suggesting the need for a temperature-aware model to find optimum elevation and a corresponding bifacial gain. Next, we explore the effect of elevation on temperature-dependent energy yield.

Ref. [47] shows that the energy yield of the module starts to

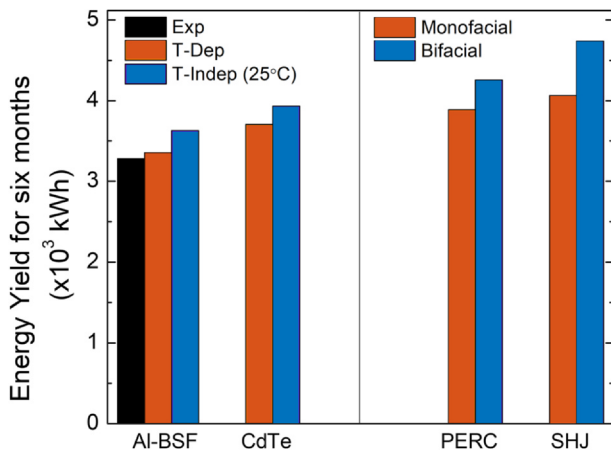


Fig. 8. Energy yield estimated for different technologies illustrating the temperature effect (left) and monofacial and bifacial effect (right) for Albuquerque, NM, USA.

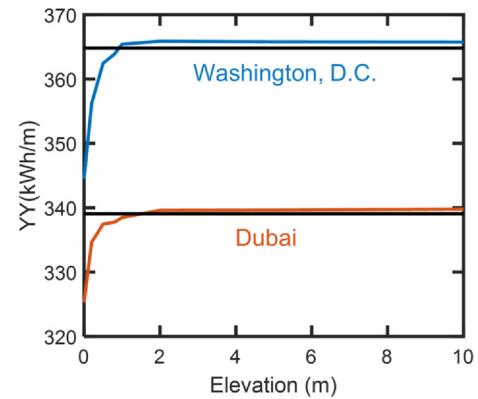


Fig. 9. Temperature – dependent yearly energy yield of the farm (with $p = 2$ m and $h = 1$ m) saturates with elevation of panels above the ground. Here we show the simulation results for Washington D.C. and Dubai, as examples.

saturate after a certain elevation. We observe a similar trend in the farm-level, i.e., yearly energy yield saturates after a certain elevation, as shown in Fig. 9. For Washington D.C., USA, YY increases from 345 kWh/m for ground-mounted panel-array ($E = 0$) and saturates to 365 kWh/m at a high elevation of panels ($E = 5$ m). At elevations $E_0 = 1.5$ m, 95% of this saturated yield can be achieved (for $h = 1$ m). Such elevation threshold, where 95% of the maximum/saturated yield would be attained is location-specific. For example, in Jeddah, KSA, this elevation threshold is $E_0 = 2$ m. We observed that this elevation-dependent saturation is observed globally. The light collection (I_{POA}) on the panels increases with elevation, leading to a rise in the temperature-dependent degradation of efficiency (see Eqs. (12)–(14) temp. models). Thus, on one hand, elevation increases light collection which enhances energy yield. On the other hand, increasing light collection simultaneously increases the temperature and decreases efficiency, eventually decreasing the energy yield. Overall, it is observed that the energy yield increases until a threshold elevation (E_0) and then saturates to a maximum value.

Fig. 10 shows the percentage change in yearly energy yield and $LCOE^*$ between an elevated bifacial farm (with $E = 2$ m) and previously estimated unelevated bifacial farm ($E = 0$ m). We chose $E = 2$ m since it is beyond the elevation threshold E_0 for energy yield saturation around the world. Evidently, the energy yield improves by 1–20% around the globe with a monotonically increasing trend with latitude. Consequently, $LCOE^*$ decreases by the same amount with a similar trend worldwide. Interestingly, the optimum tilt angle varies from -10% to $+10\%$ depending on the latitude and the fraction of diffuse light illuminating the bifacial panels. Thus, elevating the farms by 1–2 m enhances the energy yield and reduces $LCOE^*$ (assuming a negligible increase in costs). In practice, elevating the modules to $E = 2$ m incurs higher costs due to at least two factors: (1) higher wind loads require more steel and deeper foundation; (2) installing at that height increases labor costs due to the need for ladders and other work aids. To incorporate this increased cost, a higher cost ratio (M_L) in Eq. (19) should be used. Consequently, the optimum value of elevation would be lower, $1 < E_0 < 2$ m. Thus, Fig. S3 in the supplementary information shows the gain in energy yield for a practical and affordable elevation of $E = 1$ m with respect to $E = 0$ m. A detailed $LCOE$ -dependent analysis of the elevated bifacial farm design will be a part of a future analysis.

4. Summary and conclusions

In this paper, we have analyzed the effects of temperature-dependent efficiency degradation on the energy yield and $LCOE$ of monofacial and bifacial solar PV farms. Our approach involved combining an

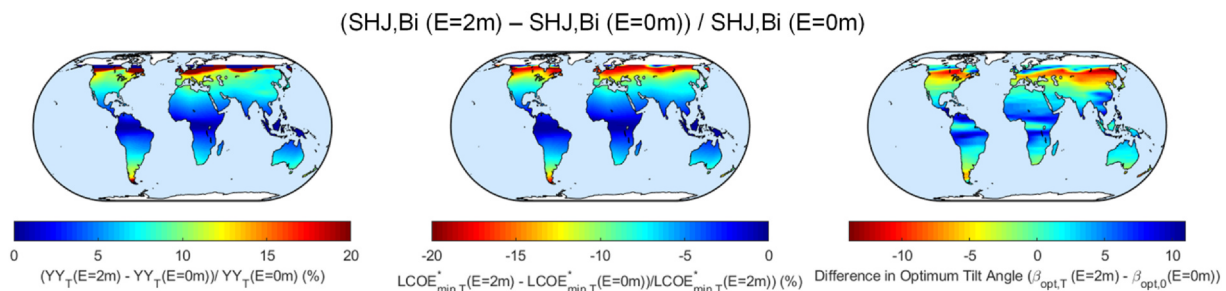


Fig. 10. Global trends in change in yearly energy yield, LCOE*, and difference in optimum tilt angle of an elevated bifacial farm ($E = 2$ m) and an unelevated bifacial farm ($E = 0$ m).

irradiance model, an updated light collection model for elevated farms, and temperature-dependent efficiency models to arrive at the final energy output of a solar farm. The light collection and temperature estimation models had to be solved self-consistently in order to arrive at the practical and more accurate efficiency for a particular location. We applied these models for locations around the world to deliver the global maps which quantify the percentage change in energy yield and $LCOE^*$ between temperature-dependent and constant temperature calculations while presenting general global trends.

Our analysis leads to the following key **conclusions**:

- The generalized Alam-Sun thermal model (Eq. 14) allows us to account for sub-bandgap absorption. The effect is significant: Almost 90 kWh more energy can be generated for $\gamma = 1$ as compared to $\gamma = 0$ from 12 bifacial-module string for a period of 6 months due to the reduced self-heating associated with the transmittance of sub-band irradiance.
- A comparison between energy yield and LCOE for temperature-dependent efficiency ($\eta(T)$) and temperature-independent efficiency (η_{STC}) predicts a change of -7% (Al-BSF), -5% (SHJ) for locations close to the equator ($|\text{Latitude}| < 30^\circ$) and $+12\%$ (Al-BSF), $+5\%$ (SHJ) for locations close to the poles ($|\text{Latitude}| > 30^\circ$).
- The bifacial gain for SHJ (bifacial) vs. traditionally used Al-BSF (monofacial) with temperature-dependent efficiency shows enhancement of $+12\%$ for hotter locations close to the equator ($|\text{Latitude}| < 30^\circ$) and can reach up to $25 - 45\%$ for colder places close to the poles ($|\text{Latitude}| > 30^\circ$). Whereas, bifacial gain for Al-BSF is limited to $0 - 30\%$. This presents an incredible opportunity for SHJ bifacial farm deployment.
- The trend in the difference in optimum tilt angle ($\Delta\beta_{opt}$) for bifacial designs is $0^\circ - 30^\circ$ when comparing the same technology (Al-BSF (Bifacial) vs. Al-BSF(Monofacial)) or two different technologies (SHJ (Bifacial) vs. Al-BSF(Monofacial)).
- Elevated farms show two counter-balancing trends, where light collection on the panels (I_{POA}) increases leading to an increase in temperature and a decrease in efficiency. Overall, temperature-dependent elevated farms ($E = 1 - 2$ m) outperform unelevated farms in terms of yearly energy yield by $1 - 20\%$ depending on the latitude.
- Bifacial PV technologies (SHJ) with lower TC and low sub-bandgap (IR) absorption can outperform their monofacial counterparts. The extent of enhancement in performance depends on the bifacial technology used and the geographical location of the farm.

In conclusion, it is important to accurately calculate the energy yield (YY) and $LCOE^*$ for any new PV technology so as to establish its economic viability. The bifacial solar farm energy yield using temperature-dependent efficiency fulfills this purpose. Although the design of the farms in terms of the optimum tilt angle is not affected significantly, the absolute values of energy yield and LCOE for field-deployed temperature-dependent solar farms differ considerably for several locations

around the world. This affects the overall economic evaluation of location-specific solar farms. Since bifacial panels have a lower temperature coefficient compared to monofacial panels, therefore, they are advantageous for relatively steady energy output due to daily and monthly temperature variations, especially for locations with lower total irradiance. Moreover, using PV technologies with lower temperature coefficients, e.g., Silicon Heterojunction (SHJ) and PERC, would lead to stable outputs and enhanced performance of bifacial solar farms throughout the world.

Finally, the scope of this study was limited to the initial temperature-dependent efficiency degradation associated with a newly installed solar farm. The long-term analysis of temperature-dependent degradation mechanisms and PV lifetime are not considered in our study. Moreover, the spectral dependence of PV efficiency is beyond the scope of this work and needs further investigation [41,48–50]. Furthermore, given the importance of TC, it will be important to focus on methods to lower the temperature coefficient of bifacial solar materials/ technologies and reduce cell temperature variation through improved packaging [51]. These are important considerations for concentrated solar photovoltaics as well. One could also improve the power output at the back surface by using a different kind of solar cell, e.g. DSSC, that performs better under low-light illumination [52–54,55]. Thus, the temperature-dependent energy-gain of a tandem [17,39,56] or ‘mixed’ bifacial solar cell will be an interesting and important generalization of the results presented in this paper.

CRediT authorship contribution statement

M. Tahir Patel: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing, Visualization. **Ramachandran A. Vijayan:** Software, Writing - original draft. **Reza Asadpour:** Investigation. **M. Varadharajaperumal:** Supervision. **M. Ryyan Khan:** Methodology, Software. **Muhammad A. Alam:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2020.115405>.

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